

Problem Set: Complex Numbers

Basic Properties and Geometric Interpretation

A Collection of Problems

Throughout, $z, w, z_1, z_2, \dots$ denote complex numbers, $\bar{z}$ the complex conjugate of z , $|z|$ the modulus, and $\arg z$ the argument. The set of complex numbers is $\mathbb{C}$, the unit circle is $S^1 = \{z : |z| = 1\}$, and $\omega = e^{2\pi i/3}$ is a primitive cube root of unity unless mentioned otherwise. Where the problem mentions a triangle in the complex plane, the vertices are labelled in counterclockwise order unless specified.

Part 1 — Algebraic Manipulation and Basic Identities

- Find all complex numbers $z \neq 0$ such that $z + \frac{1}{z}$ is real.
- Compute:
 - $i^{2000} + i^{1999} + i^{201} + i^{82} + i^{47}$.
 - $1 + i + i^2 + i^3 + \dots + i^n$ for $n = 100, 101, 102, 103$.
 - $i^1 \cdot i^2 \cdot i^3 \dots i^{2021}$.
 - $i^{-5} + (-i)^{-7} + (-i)^{13} + i^{-100} + (-i)^{94}$.
- Let z be a nonzero complex number such that $\left|z^3 + \frac{1}{z^3}\right| \leq 2$. Prove that $\left|z + \frac{1}{z}\right| \leq 2$.
- Consider z a complex number with $\operatorname{Re}(z) > 1$. Prove that $\left|\frac{1}{z} - \frac{1}{2}\right| < \frac{1}{2}$.
- Let z_1, z_2 be distinct complex numbers such that $|z_1| = |z_2|$. Prove that $\frac{1}{2}|z_1 + z_2| < |z_1|$.
- Let $n > 2$ be an integer. Find the number of solutions to the equation $z^{n-1} = i\bar{z}$.
- Let $z^2 + z + 1 = 0$. Find the value of

$$\left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \dots + \left(z^{12} + \frac{1}{z^{12}}\right)^2.$$
- Let z_1, z_2, z_3 be complex numbers such that $z_1 + z_2 + z_3 = 0$ and $|z_1| = |z_2| = |z_3| = 1$. Prove that $z_1^2 + z_2^2 + z_3^2 = 0$.
- Let z_1, z_2, z_3 denote the vertices of an equilateral triangle in the complex plane, and let z_0 be the centroid of the triangle. Prove that $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$.
- Suppose real numbers x, y satisfy the equations

$$x + \frac{x + 8y}{x^2 + y^2} = 2, \quad y + \frac{8x - y}{x^2 + y^2} = 0,$$

simultaneously. Find the value(s) of x and y .

11. Let z_1, z_2, z_3 denote the vertices of a right-angled isosceles triangle, right-angled at z_2 . Prove that $z_1^2 + 2z_2^2 + z_3^2 = 2z_2(z_1 + z_3)$.

12. Let z_1, z_2, z_3 be distinct complex numbers such that $|z_1| = |z_2| = |z_3| > 0$. If $z_1 + z_2z_3$, $z_2 + z_1z_3$ and $z_3 + z_1z_2$ are real numbers, prove that $z_1z_2z_3 = 1$.

13. Let z_1, z_2 be complex numbers with $|z_1| = |z_2| = 1$. Prove that

$$|z_1 + 1| + |z_2 + 1| + |z_1z_2 + 1| \geq 2.$$

14. Let z be a complex number such that $\text{Im}(z) \neq 0$, and $\frac{1+z+z^2}{1-z+z^2}$ is real. Prove that $|z| = 1$.

15. Let z_1, z_2, z_3 be complex numbers with $z_1 + z_2 + z_3 = z_1z_2 + z_2z_3 + z_3z_1 = 0$. Prove that $|z_1| = |z_2| = |z_3|$.

Part 2 — Modulus, Conjugate and the Unit Circle

16. Let $z, w \in \mathbb{C}$. Prove the *parallelogram identity*

$$|z + w|^2 + |z - w|^2 = 2(|z|^2 + |w|^2),$$

and use it to show that for any four points a, b, c, d in the plane, the sum of the squares of the diagonals of the quadrilateral $abcd$ equals the sum of the squares of the four sides minus four times the square of the distance between the midpoints of the diagonals.

17. Show that for any $z \in \mathbb{C}$,

$$|1 + z| + |1 - z| \geq 2,$$

with equality iff z is real and $|z| \leq 1$. Generalise: for any fixed $w \neq 0$, find $\inf_{|z|=1} (|w + z| + |w - z|)$.

18. Prove that for any $z \in \mathbb{C}$ with $|z| = 1$ and $z \neq -1$, the number $\frac{1+z}{1-z} \cdot \frac{1}{i}$ is real, and find the corresponding statement for z with $|z| = r$.

19. For complex z, w with $|z| < 1$ and $|w| < 1$, prove that $\left| \frac{z-w}{1-\bar{z}w} \right| < 1$, with equality on the boundary $|z| = 1$ or $|w| = 1$. (*Schwarz–Pick lemma, simplified form.*)

20. Show that for z, w on the unit circle with $zw \neq 1$,

$$\left| \frac{z-w}{1-zw} \right| = 1.$$

Use this to characterise all Möbius transformations of the unit disk to itself.

21. Let $a, b, c \in \mathbb{C}$ with $|a| = |b| = |c| = 1$. Prove

$$|a - b| \cdot |b - c| \cdot |c - a| \leq 3\sqrt{3},$$

and identify the equality case.

22. For complex numbers $z_1, \dots, z_n$, prove the *triangle inequality with refinement*:

$$||z_1| - |z_2| - \dots - |z_n|| \leq |z_1 + z_2 + \dots + z_n| \leq |z_1| + |z_2| + \dots + |z_n|,$$

and characterise when each inequality becomes equality.

23. Let $z_1, z_2, \dots, z_n$ be complex numbers with $|z_k| \leq 1$ for all k . Prove that there exist $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n \in \{-1, +1\}$ such that

$$\left| \sum_{k=1}^n \varepsilon_k z_k \right| \leq \sqrt{2}.$$

(Hint: pair complex numbers by argument.)

Part 3 — Roots of Unity and Polynomial Identities

24. Let $\omega = e^{2\pi i/n}$ be a primitive n th root of unity. Prove

$$1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0, \quad \prod_{k=1}^{n-1} (1 - \omega^k) = n.$$

25. Let $\omega = e^{2\pi i/7}$. Compute

$$S_1 = \omega + \omega^2 + \omega^4, \quad S_2 = \omega^3 + \omega^5 + \omega^6,$$

and use $S_1 + S_2$ and $S_1 S_2$ to deduce S_1, S_2 explicitly. Hence show $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} = -\frac{1}{2}$.

26. Let $\omega = e^{2\pi i/n}$. Prove that for any integer k with $1 \leq k \leq n-1$,

$$\sum_{j=0}^{n-1} \omega^{jk} = 0.$$

Apply this to evaluate $\sum_{j=0}^{n-1} \cos \frac{2\pi jk}{n}$ and $\sum_{j=0}^{n-1} \sin \frac{2\pi jk}{n}$.

27. Let $z_1, z_2, \dots, z_n$ be the n th roots of a complex number $a \neq 0$. Prove that

$$z_1 z_2 \cdots z_n = (-1)^{n-1} a, \quad \sum_{k=1}^n z_k = 0 \text{ for } n \geq 2.$$

28. Find all $z \in \mathbb{C}$ satisfying $z^7 = \bar{z}$. Describe the solution set geometrically.

29. Solve $(z+1)^n = (z-1)^n$ over $\mathbb{C}$, for an integer $n \geq 1$. Show that all solutions are pure imaginary, and express them as $z_k = i \cot(k\pi/n)$ for $k = 1, 2, \dots, n-1$.

30. Prove that for any positive integer n and any complex z ,

$$\prod_{k=0}^{n-1} \left(z - e^{2\pi i k/n} \right) = z^n - 1.$$

Setting $z = -1$, deduce $\prod_{k=1}^{n-1} \sin \frac{k\pi}{n} = \frac{n}{2^{n-1}}$.

31. Let $\omega \neq 1$ be an n th root of unity. Prove

$$1 + 2\omega + 3\omega^2 + \dots + n\omega^{n-1} = \frac{n}{\omega - 1}.$$

32. Let $P(z) = z^n + a_{n-1}z^{n-1} + \dots + a_0$ have all roots on the unit circle. Prove that $|a_0| = 1$ and that the coefficients of P are determined by the roots up to a unimodular constant. Conversely, give an example of a degree- n monic polynomial with $|a_0| = 1$ but not all roots on the unit circle.

Part 4 — Geometry in the Complex Plane

33. Three distinct points z_1, z_2, z_3 in the complex plane are collinear if and only if $\frac{z_3 - z_1}{z_2 - z_1}$ is real. Prove this. State the analogous criterion for z_1, z_2, z_3, z_4 to be concyclic (or collinear).

34. Three points z_1, z_2, z_3 form an equilateral triangle (in counterclockwise order) iff

$$z_1 + \omega z_2 + \omega^2 z_3 = 0,$$

where $\omega = e^{2\pi i/3}$. Prove this and prove the equivalent condition $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$.

35. The triangle with vertices z_1, z_2, z_3 is similar (with the same orientation) to the triangle with vertices w_1, w_2, w_3 iff

$$\det \begin{pmatrix} z_1 & w_1 & 1 \\ z_2 & w_2 & 1 \\ z_3 & w_3 & 1 \end{pmatrix} = 0.$$

Prove this and state the corresponding criterion for opposite (reflected) similarity.

36. Given $\triangle ABC$ with vertices $a, b, c \in \mathbb{C}$, prove the centroid is $\frac{a+b+c}{3}$, and the circumcentre is the unique solution o of $|o-a|^2 = |o-b|^2 = |o-c|^2$. Derive the explicit formula

$$o = \frac{|a|^2(b-c) + |b|^2(c-a) + |c|^2(a-b)}{a(\bar{b}-\bar{c}) + b(\bar{c}-\bar{a}) + c(\bar{a}-\bar{b})}.$$

37. Prove that the orthocentre of a triangle with vertices a, b, c on the unit circle is $h = a+b+c$. Hence prove the Euler line: the centroid g , circumcentre o , and orthocentre h of any triangle satisfy $o\vec{h} = 3 \cdot o\vec{g}$, i.e. $h = 3g - 2o$.

38. For three points a, b, c on the unit circle, prove that the foot of the altitude from a to bc has the complex coordinate $\frac{1}{2}(a+b+c-bc/a)$. Use this to give a one-line proof that the orthocentre is $a+b+c$.

39. $ABCDEF$ is a hexagon inscribed in the unit circle, with vertices $a, b, c, d, e, f \in S^1$. Prove that $ABCDEF$ has parallel opposite sides ($AB \parallel DE$, etc.) iff $abc = def$. Use this to derive a complex-analytic proof of one direction of Pascal's theorem.

40. (*Ptolemy via complex numbers*) For four complex numbers a, b, c, d , prove

$$|(a-c)(b-d)| \leq |(a-b)(c-d)| + |(a-d)(b-c)|,$$

with equality iff a, b, c, d are concyclic in this cyclic order. Deduce the classical Ptolemy identity for cyclic quadrilaterals.

41. Show that the Möbius transformation $f(z) = \frac{az+b}{cz+d}$ (with $ad-bc \neq 0$) maps the family of circles and lines to itself. Use this to prove that the cross-ratio

$$[z_1, z_2; z_3, z_4] = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}$$

is invariant under Möbius transformations.

42. Let z_1, z_2, z_3, z_4 be four points on the unit circle. Prove

$$[z_1, z_2; z_3, z_4] \in \mathbb{R} \iff z_1, z_2, z_3, z_4 \text{ are concyclic (or collinear)}.$$

Combining with the Möbius invariance of cross-ratio, deduce that any three points can be mapped to any three given points by a unique Möbius transformation.

Part 5 — Inequalities, Sums and Advanced Identities

43. Prove the *Lagrange identity* for complex numbers:

$$\left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 = \left(\sum_{k=1}^n |a_k|^2 \right) \left(\sum_{k=1}^n |b_k|^2 \right) - \sum_{1 \leq j < k \leq n} |a_j \bar{b}_k - a_k \bar{b}_j|^2.$$

Deduce the Cauchy–Schwarz inequality.

44. For $n \geq 1$ and complex $z_1, \dots, z_n$ with $|z_k| \leq 1$ for all k , prove

$$\left| \sum_{k=1}^n z_k \right|^2 + \sum_{1 \leq j < k \leq n} |z_j - z_k|^2 \leq n^2.$$

45. Let $a_0, a_1, \dots, a_n$ be complex numbers, and define $P(z) = a_0 + a_1 z + \dots + a_n z^n$. Prove

$$\frac{1}{2\pi} \int_0^{2\pi} |P(e^{i\theta})|^2 d\theta = \sum_{k=0}^n |a_k|^2.$$

(Parseval's identity for polynomials.)

46. (*Schur's inequality, complex version*) Let z_1, z_2, z_3 be complex numbers with $|z_k| = 1$ for $k = 1, 2, 3$. Prove that

$$|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2 \leq 9,$$

with equality iff z_1, z_2, z_3 are the vertices of an equilateral triangle inscribed in S^1 .

47. Let $z_1, z_2, \dots, z_n$ be complex numbers. Prove

$$\sum_{1 \leq j < k \leq n} |z_j - z_k|^2 = n \sum_{k=1}^n |z_k|^2 - \left| \sum_{k=1}^n z_k \right|^2.$$

Apply this to give a one-line proof that the centroid g of n points minimises $\sum_k |z - z_k|^2$.

48. Let $z_1, z_2, \dots, z_n$ be on the unit circle, and let $w \in \mathbb{C}$ with $|w| < 1$. Prove

$$\sum_{k=1}^n \left| \frac{z_k - w}{1 - \bar{w} z_k} \right|^2 = n.$$

49. Find the maximum of $|z^3 - z + 2|$ over the closed unit disk $|z| \leq 1$. (*Hint: by the maximum principle, the maximum is attained on the boundary $|z| = 1$.*)

50. (*Concluding identity*) Let $z_1, z_2, \dots, z_n$ be the n th roots of unity. Prove that for any complex number w ,

$$\prod_{k=1}^n (w - z_k) = w^n - 1, \quad \sum_{k=1}^n \frac{1}{w - z_k} = \frac{nw^{n-1}}{w^n - 1},$$

and apply the second identity with $w = 2$ to evaluate

$$\sum_{k=1}^n \frac{1}{2 - e^{2\pi i k/n}}.$$